RootSync

RCO / Tetherless

Smartphone Compatibility Baseline & Virtual Testing Strategy

Combined and reconciled research for 2–3 GB RAM, ~5 GB free storage, and DDIL/offline operation

Executive conclusion

The two research streams mostly agree on the device classes and testing tools, but several support-floor claims should be tightened. For Tetherless, RootSync should use iOS 16.4+ and Android 10+ as the official minimum support baseline, with 2 GB RAM as the stress-test floor, 3 GB preferred, and at least 5 GB free storage at the time of use. iOS 15 / iPhone 7 and Android 9 can be treated as legacy best-effort targets only if RCO's actual device inventory shows they are common. Android 8.1 should not be an official baseline because current Chrome no longer updates there. Testing should combine emulators/simulators with real-device testing; no virtual environment alone can validate low-memory performance or multi-day offline persistence.

1. Purpose and RCO constraints

This report combines RootSync's earlier device research with the additional smartphone-baseline research supplied by the team and reconciles both against current official platform documentation.

- Tetherless must support DDIL conditions: disconnected, disrupted, intermittent, or limited connectivity.
- RCO expects some learners to use older or budget smartphones with roughly 2–3 GB RAM.
- RCO indicated users may have around 5 GB of free storage available.
- The app is intended to be lightweight, mobile-first, and usable without continuous internet access.
- RootSync is currently favoring a React + TypeScript + Vite PWA with Service Worker caching and Dexie/IndexedDB for local data.
- The device baseline must be realistic enough for RCO's users while avoiding support for browsers that no longer receive security updates.

2. Does the new research match our earlier research?

Mostly yes. The phone-model examples, RAM tiers, and testing-tool choices are directionally consistent. The main differences are the recommended OS floors and several storage/PWA claims that need qualification.

Claim from added research	Assessment	Reconciled conclusion	Reason
iOS 15+ (or 16+) baseline	Partly matches	Use iOS 16.4+ officially; iOS 15 as legacy best-effort	Current Vite defaults target Safari 16.4+, reducing compatibility risk.
iPhone 8 / 7 as low-end hardware	Good stress-test examples	iPhone 8 is the official minimum representative; iPhone 7 only if RCO needs legacy support	iPhone 8 receives iOS 16 security updates; iPhone 7 remains on iOS 15.
Android 8.1/9/10 baseline	Too broad / outdated as official floor	Use Android 10+ officially; Android 9 best-effort only	Chrome 139+ requires Android 10; Android 8/9 are frozen on Chrome 138.
32/64 GB phones will have 5–10 GB free	Not guaranteed	Require ≥5 GB free storage at runtime	Free space depends on user apps, photos, cached media, and OS usage—not factory capacity alone.

Home Screen PWA avoids Safari 7-day ITP storage deletion	Largely correct	Installed Home Screen web app avoids ITP's 7-day first-party purge, but storage can still be evicted under storage pressure	WebKit explicitly exempts Home Screen web-app first-party data from the 7-day cap, but quota/pressure eviction still exists.
Android 8.1+ has unrestricted Service Worker/IndexedDB/background sync	Overstated	Service Worker/IndexedDB are mature, but Background Sync should not be a cross-platform requirement	Old Android Chrome is no longer current; sync should also run on app start/resume/online/manual action.
Android Studio, DevTools, BrowserStack, Xcode Simulator are appropriate	Matches	Keep all four, but use them for different layers of testing	Emulators/simulators do not fully reproduce physical device performance or long-term storage behavior.

3. Recommended lowest common denominator for Tetherless

Support level	iPhone / iOS	Android	RAM	Storage
Official minimum	iOS 16.4+; representative minimum iPhone 8 / iPhone X class	Android 10+ with supported/current Chrome/WebView	2 GB stress-test floor	≥5 GB free at time of use
Preferred low-end target	iPhone SE 2nd gen or newer; 3 GB-class hardware	Android 11/12 low-end phone; 3 GB RAM	3 GB preferred	≥5 GB free
Legacy best effort	iOS 15.8.x / iPhone 7 only if inventory requires it	Android 9 / Chrome 138 only if inventory requires it	2 GB	≥5 GB free
Not promised by default	iOS 14 and older	Android 8.1 and older	<2 GB	<5 GB free

Recommended PRD wording: “Tetherless officially targets iOS 16.4+ and Android 10+, with 2 GB RAM as the minimum stress-test class, 3 GB preferred, and at least 5 GB of free device storage. Older OS versions may be evaluated only if RCO device inventory shows material usage.”

4. iPhone baseline and representative models

Apple does not publish RAM figures in its normal consumer specifications, so RAM numbers for older iPhone models are typically taken from third-party hardware references. The OS-support limits below are supported by Apple documentation.

Device	Approx. RAM class*	OS position	RootSync status	Use in test plan
iPhone 7	~2 GB	iOS 15.8.x family	Legacy best effort	Test only if RCO inventory shows meaningful use
iPhone 8 / 8 Plus	~2/3 GB	iOS 16.7.x family	Recommended minimum representative	Primary iOS stress-test class
iPhone X	~3 GB	iOS 16.7.x family	Supported	Good 3 GB-class iOS reference
iPhone XR / XS	~3/4 GB	Newer iOS support than iPhone 8/X	Supported	Secondary compatibility test
iPhone SE 2nd gen (2020)	~3 GB	Still in newer iOS support line	Preferred low-end reference	Best practical low-end iPhone target

* RAM figures are approximate third-party hardware specifications; Apple’s public device pages generally do not list RAM.

Why iOS 16.4+ is the recommended floor: current Vite production builds target Safari 16.4+ by default. Supporting iOS 15 is technically possible, but it increases compatibility/polyfill testing and should be justified by RCO’s actual device population.

5. Android baseline and representative models

Android has a wider hardware range, so the OS/browser floor is more important than a single model. Google ended new Chrome releases for Android 8 and Android 9 after Chrome 138; Chrome 139+ requires Android 10.

Representative phone	RAM	OS	RootSync role	Notes

Nokia 2.4	2 / 3 GB	Android 10	Minimum Android reference	Official HMD specs closely match RCO's lower-bound requirement
Nokia C21 Plus	2 / 3 GB	Android 11 Go	Low-end / Go-class stress test	Useful for constrained-memory and low-cost-device testing
Moto e22	3 / 4 GB	Android 12	Preferred low-end reference	Motorola lists 3/4 GB RAM and Android 12
Typical Android 9 budget phone	2–3 GB	Android 9 / Chrome 138 max	Legacy best effort	May function, but browser no longer receives current Chrome updates
Android 8.1 budget phone	2–3 GB	Android 8.1 / Chrome 138 max	Not official support	Avoid unless RCO proves a strong need

6. How to interpret RCO's "up to 5 GB available storage" requirement

The device's factory storage capacity (32 GB, 64 GB, etc.) does not tell us how much space an athlete actually has left. A 32 GB phone can have 15 GB free or almost none depending on photos, apps, and cached data.

- Treat "5 GB free" as a runtime acceptance condition, not as a model specification.
- The Tetherless learner app should remain far below that limit; the MVP should target a cached footprint measured in tens or low hundreds of MB, not gigabytes.
- Training event records are tiny compared with images/video; avoid large offline video packages unless RCO specifically requires them.
- Use navigator.storage.estimate() where available to monitor origin storage usage/quota, but do not assume browser storage is permanent.
- IndexedDB and Cache Storage are best-effort by default and may be evicted under device storage pressure.
- Home Screen installation on iOS protects the first-party app from WebKit's ITP 7-day script-storage purge, but does not eliminate storage-pressure or quota-based eviction.

7. PWA / IndexedDB implications by platform

iOS

- Service Workers and IndexedDB are mature enough for the proposed offline architecture on iOS 16.4+.
- A true Home Screen web app is preferable to relying only on a Safari tab for long-lived offline use.
- WebKit explicitly exempts Home Screen web-app first-party domains from ITP's 7-day script-writable storage cap.
- Storage can still be removed under storage pressure or broader quota rules, so unsynced learner events must be kept small and synchronized whenever possible.
- Do not claim that installing the PWA makes data impossible to lose; real-device low-storage tests remain necessary.

Android

- Android 10+ with current Chrome provides a strong PWA, Service Worker, Cache Storage, and IndexedDB environment.
- Android 8/9 may still run Chrome 138, but they no longer receive current Chrome releases and should not define the official baseline for a cybersecurity organization.
- Background Sync may be available in Chrome, but Tetherless should not depend on it. Sync should also trigger on app launch/resume, the online event, and a manual Sync action.

8. Virtual and real-device test environments

Tool	Best use	What it can model	Important limitation
Android Studio AVD	Functional Android compatibility and constrained-device profiles	Android API level, RAM, internal storage, screen resolution, network latency, device profile	Does not perfectly reproduce a cheap physical phone's CPU, thermal, battery, and memory-pressure behavior
Chrome DevTools	Daily development stress testing	CPU throttling, custom network speed/latency, offline mode, storage quota simulation, Service Worker/IndexedDB inspection	Approximation only; not a real 2 GB phone
Xcode Simulator + Safari Web Inspector	iOS/Safari/PWA functional compatibility	Different simulator runtimes/devices; web pages, Service Workers, Home Screen web apps, IndexedDB debugging	Simulator does not reproduce physical-device performance or real RAM/storage pressure; not every historical iOS/device combination is always available
BrowserStack Real Device Cloud	Cross-device validation on physical iOS/Android hardware	Real Safari/Chrome, real touch/OS behavior, many device/OS combinations	Sessions/devices are reset; not ideal for multi-day persistence tests, and plan availability varies
Owned physical low-end phones	Final DDIL/performance validation	Real memory pressure, storage pressure, app restarts, days offline, actual touch/CPU behavior	Requires acquiring/borrowing a small device set

9. Recommended virtual device profiles for RootSync

Android Studio AVD profiles

- **RCO-Minimum-Android10:** Android 10 / API 29, 2 GB RAM, 720×1600-class screen, 32 GB internal storage, test with only ~5 GB free, current compatible Chrome image where possible.
- **RCO-Go-Android11:** Android 11 Go-style profile, 2 GB RAM, 720p-class screen, constrained storage, slow network profile.
- **RCO-Preferred-Android12:** Android 12, 3 GB RAM, 720p-class screen, 32/64 GB internal storage, representative low-end target.

iOS simulator profiles

- Use iOS 16.x simulator runtime as the minimum functional web-platform baseline where available.
- Also test a current iOS simulator to catch forward-compatibility issues.
- Use Safari Web Inspector to inspect Service Workers, Home Screen web apps, storage, and JavaScript behavior.
- Do not use simulator results as proof of 2 GB RAM performance; validate iPhone 8/SE-class hardware on a real device or real-device cloud.

10. Recommended physical device matrix

Priority	Device class	Representative model	Target OS	Purpose
P0	2 GB iPhone stress test	iPhone 8	iOS 16.7.x	Minimum iOS performance/offline persistence
P0	2 GB Android stress test	Nokia 2.4	Android 10	Minimum Android performance/offline persistence
P1	3 GB iPhone reference	iPhone SE 2nd gen / iPhone X-class	Supported iOS	Preferred low-end iOS experience
P1	2–3 GB Android Go reference	Nokia C21 Plus	Android 11 Go	Constrained Android behavior
P1	3 GB newer low-end Android	Moto e22	Android 12	Representative newer budget phone
P2 legacy	Old iOS	iPhone 7	iOS 15.8.x	Only if RCO inventory requires

				iOS 15 support
P2 legacy	Old Android	Android 9 budget device	Android 9 / Chrome 138	Only if RCO inventory requires legacy Android support

11. Required DDIL test sequence on every minimum device profile

1. Install/open Tetherless while online and download the assigned module.
2. Confirm app/content footprint and available storage.
3. Disable Wi-Fi/cellular connectivity.
4. Close and reopen the installed app while offline.
5. Complete several exercises and verify IndexedDB/Dexie writes succeed.
6. Close/reopen again and confirm all progress remains.
7. Keep at least one physical test device offline for the agreed multi-day period.
8. Reconnect using a slow/unstable network profile.
9. Interrupt synchronization midway, reconnect, and verify retry/resume behavior.
10. Repeat the same sync and confirm no duplicate backend events are created.
11. Verify the instructor/reporting side matches accepted backend events.
12. Test low-storage behavior and verify the app fails safely if a storage quota is reached.

12. Compatibility decision gates

- PASS: iPhone 8 / iOS 16.7 and Nokia 2.4 / Android 10 can open the installed app offline, complete training, preserve data after restart, and later sync without duplicates.
- PASS: the app remains usable with 2 GB-class hardware and 5 GB free storage without excessive startup/load times.
- FAIL: IndexedDB data disappears during normal offline/restart tests, storage-pressure testing causes unacceptable loss, or sync cannot recover reliably.
- If browser/PWA storage is the failure point rather than application logic, escalate to the previously proposed Capacitor + native SQLite hardening path.

13. One data request that will remove most uncertainty

Ask RCO for a small anonymous device inventory from roughly 10–20 representative cyber athletes. No personal data is necessary. A simple list is enough:

Phone model | OS version | approximate free storage

This allows RootSync to replace assumptions with an exact support matrix and decide whether iOS 15 / Android 9 legacy support is actually worth the engineering cost.

14. Final recommendation

Recommended official baseline

iOS 16.4+ / iPhone 8-class or newer
Android 10+ / 2 GB RAM or more
3 GB RAM preferred on both platforms
At least 5 GB free storage at time of use
Installed PWA / Home Screen use encouraged
Real-device validation required before final architecture lock

This baseline is slightly stricter than the added research but is safer for a cybersecurity training product because it aligns with current browser support and avoids making obsolete, non-updating browsers part of the official support promise. If RCO's actual device survey shows substantial iPhone 7 / Android 9 usage, RootSync can add a legacy-support tier and configure/test the build accordingly.

15. Key sources

- Vite - Building for Production / browser compatibility
- Google Chrome - Android 8/9 support sunset announcement
- Apple - iOS 16 compatible devices
- Apple - Security releases (shows ongoing iOS 16/15 device lines)
- WebKit - Tracking Prevention / Home Screen web-app exemption
- WebKit - Storage policy and eviction
- MDN - Storage quotas and eviction criteria
- Android Developers - Create and manage virtual devices
- Chrome DevTools - Throttling
- Chrome DevTools - Application panel
- Apple Developer - Running on simulated or physical devices
- Apple Developer - Inspecting iOS/iPadOS web content
- BrowserStack - Real iOS/Android devices
- HMD - Nokia 2.4 specifications
- HMD - Nokia C21 Plus specifications
- Motorola - moto e22 specifications

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